

ZANZIBAR MALARIA ELIMINATION PROGRAMME



ANNUAL REPORT
July 2021 / June 2022



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Abbreviations and Acronyms

Abbreviation	Definition
ABER	Annual blood examination rate
ACTs	Artemisinin based combination therapies
Ai/ m ²	Active Ingredient per meter square
ALU	Artemether Lumefantrine.
ANC	Ant Natal Care
ASAQ	Artesunate plus amodiaquine.
bti	Bacillus thuringiensis israelensis
C19RM	COVID-19 Response Mechanism
CAG	Controller and Auditor General
CHV	Community Health Volunteers
CMS	Central medical store.
CPO	Chief Pharmacist Office
DHIS2	District Health Information System version 2
DMO	District medical officer.
DMSO	District Malaria Surveillance Officers
DP	District pharmacist.
DRTs	District response teams
EBS	Event-Based Surveillance
eLMIS	Electronic Logistic Management Information System
EOC	Emergency Operation Center
FY	Financial Year
GCMS	Gas Chromatology Mass Spectrophotometer.
GF	Global Fund
GFATM Malaria	The Global Fund to Fight AIDS, Tuberculosis and Malaria



HCF	Health Care Facilities
HCWs	Health Care workers
HF's	Health Facilities
HMIS	Health Management Information System
HPU	Health Promotion Unit
IAS	International Accounting Standards
IASB	International Accounting Standards Board
IBS	Indicator-based surveillance
IDSR	Integrated Disease Surveillance and Response
IFRS	International Financial Reporting Standards
IHR	International Health Regulations
IMS	Incident Management System
IPSAS	International Public-Sector Accounting Standards
IPV	Interpersonal Violence Prevention
IRS	Indoor Residual Spraying
ITN	Insecticide-treated net
IVM	Integrated Vector Management
JnA	Jamii ni Afya app
LLIN's	Long Lasting insecticide treated nets
LMU	Logistic Management Unit
LSM	Larval Source Management
MBS	Malaria Behavior Survey
MCA	Multiple corresponding analysis
MCN	Malaria Case Register
MDA	Mass Drug Administration
MEEDs	Malaria Early Epidemic Detection System
mMQAQC	Malaria Microscope quality assurance quality control



MoH	Ministry of Health
MRC	Mass replacement campaign
mRDT	Malaria Rapid Diagnosis Test
OMDM	Okoa Maisha Dhibiti Malaria
p/μl	Parasite per microliter
PBO	Piperonyl Butoxide
Pf	Plasmodium falciparum
PHCU	Primary health Care Unit
Pm	Plasmodium malaria
PMI	Presidential Malaria Initiative
Po	Plasmodium ovale
PPE	Personal Protective Equipment
PPM	Pool Procurement Mechanism
PR	Principal Recipients
PT	Proficiency Test
Pv	Plasmodium vivax
RCD	Reactive case detection
RGOZ	Revolutionary Government of Zanzibar
RSSH	Resilient and Sustainable Systems for Health
RTI	Research Triangle Institute
SBCC	Social and behavior change
SITREP	Situation Report
SLD	Single low dose.
SR's	Sub Recipient's
TES	Therapeutic efficacy study.
TOT	Training of Trainers
US\$	United States dollar



USAID	United States Agency for International Development
WHO	World Health Organization
ZAMEP	Zanzibar Malaria Elimination Programme
ZFDA	Zanzibar Food and Drugs Agency
ZPHEOC	Zanzibar Public Health Emergency Operation Centre



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A handwritten signature in blue ink, appearing to read 'Mohamed Ali Ali'.

Mohamed Ali Ali

Program Manager

Zanzibar Malaria Elimination Program



Executive Summary

This malaria annual report 2021/2022 aims at providing an update of the malaria situation as well as attainment of targets as per annual work plan. In this context, the Programme is responsible for providing technical guidance in order to achieve malaria elimination through the use of recommended package of interventions in order to interrupt local transmission. Findings from the field reveals that vector surveillance continue to document efficiency of *Anopheles arabiensis* being a predominant malaria vector in both Islands of Zanzibar comprising of 93% with exhibiting exophagic behavior by 75%. The average vector infection rate was 0.54%, higher in Unguja (0.68%) than in Pemba (0.4%).

In an effort to reduce infection rate, field officers investigated 99% (n=99) out of 100 active malaria foci 100 and conducted a reactive focal IRS in 64% foci areas (n = 63) covering 98% of targeted structures (n =21,470), and thus protected 98% of the population (n=130,016). Additionally, insecticidal treated nets distribution is maintained through antenatal clinics and mass replacement campaign after every three years. The last campaign was conducted in September 2021 with good coverage of 96% to all 335 targeted shehia. Furthermore, a total of 99,789 have been issued in 2021 to pregnant women and children at the age of nine years in all districts.

In this reporting period, a total of 5357 confirmed malaria cases were reported from health facilities (public& private) and at community level. Out of reported malaria cases 92% (4931) were fully investigated and classified accordingly. Eighty nine percent 79.8%(3938) of investigated cases were imported and 20%(989) were indigenous by origin. The programme noted a decline of malaria cases by 42.2% compared with last fiscal year 2020/2021.

Quality of malaria diagnosis is regularly monitored, and the overall sensitivity was 96.4% and specificity was 99.9%. *Plasmodium falciparum* remains the predominant species by 89.1% (873) followed by mixed infection of *Plasmodium falciparum* and *Plasmodium malariae* 5.5% (54) *Plasmodium malariae* 4.3% (42) and *Plasmodium ovale* is 1.1% (11). No *Plasmodium vivax* documented. With regards to malaria rapid test quality management, the results revealed that the quality of rapid test was >90%. No invalid device was detected. The overall annual blood examination rate (ABER) was 30% slightly increased compared to the 27.15% of 2020/2021, of which 40% reached in Unguja and 14% in Pemba. The program continues to provide technical professional knowledge to health care providers in public and private health facilities. The training is followed by supportive supervision and monitoring conducted in 180 (75%) health facilities in Zanzibar.



Social behaviour change is given outmost importance to the general population. Malaria behaviour study of 2021 indicated an increase on utilization of malaria key interventions. For instance, insecticidal net utilization among community members is 85% and knowledge about malaria prevention is promising up to 94%. Partnership relation and coordination is well established by ZAMEP through the involvement of different health sector programmes and units, line ministries, research as part of health system strengthening.



Introduction

An annual report provides transparency and accountability and build confidence in the programme. It justifies the need for malaria elimination and the importance of increasing and sustaining domestic and international financing for the malaria elimination.

This annual malaria report was developed with the aim of reporting program performance and challenges in line with the current malaria strategic plan 2017/18 – 2022/2023. On a yearly basis, an annual report is prepared as an advocacy tool to the Government, development partners, stakeholders including research institutions in order to make a case for malaria elimination in Zanzibar.

This report outlines the progress made in all key thematic areas namely surveillance, monitoring and evaluation, malaria diagnosis and overall case management, integrated vector control, socio behavioural change communication, malaria financing and health system strengthening.

In this reporting period, a total of 88 activities were planned from different thematic areas in the 2021/2022 annual plan based on the current strategic plan. Total of 85 activities have been implemented with an overall performance of 97.2%. Details of implementation are found in the specific thematic report in the main report.

Broad objective: The overall objective of this report is to provide a progress and performance of the main malaria works based on the annual malaria plan for 2021/22 and the National Malaria Strategic Plan - 2018-2023.

Specific Objectives

- To provide information on program activities and performance regards no. of people protected and life's saved from malaria for the year
- To review the access, coverage and quality of malaria service delivery by different thematic areas of the program and identify priorities for the next planning cycle
- To give a situation analysis of the progress and performance of the program and its future direction
- To contribute to the reporting of the overall Ministry of Health annual report
- To report to all stakeholder and partners to continue their engagement and as part of good governance
- To provide advocacy and marketing tool to influence, give direction and mobilize and sustain investments.



Sources of Information

- Health information system
- Integrated disease surveillance system
- Malaria surveillance system
- Malaria Case based surveillance for elimination
- Sentinel surveillance
- Health and malaria surveys
- Operational research studies

Process and Methods Used

ZAMEP management facilitated the entire writing process by inviting senior staff within and outside the program. Quarterly reports from each quarter were used to inform the writing process. Verification of information document was often done by consulting key references available within the program and at the Ministry of Health. Thematic working teams produced draft report and presented for finalisation to the editing team.



Malaria Epidemiology in Zanzibar

The archipelago of Zanzibar is located between longitudes 39.19793 and latitudes -6.16394, 25 kilometers (km)–50km off the east coast of Mainland Tanzania in the Indian Ocean.

There are two main islands, Pemba and Unguja, which cover a total land area of 2,461 km² and have a total estimated population of 1,717,608 people. Zanzibar comprises 11 districts, which are subdivided into 387 shehias, 258 of which are on Unguja and 129 are on Pemba. Shehias are akin to wards and are Zanzibar's lowest administrative unit where many of the public services are planned, managed, and implemented, including for health and malaria. Zanzibar's climate is characterized by two main rainfall seasons: a primary season (March– May, called masika) and a secondary season (November–January, called vuli); rainfall is at its lowest in July. Both rainfall seasons are followed by peak malaria transmission seasons, with the highest malaria case count typically observed in the March–May rainfall season.

Transmission of malaria in Zanzibar is still at very low level with prevalence of less than 1% since 2017. The last community survey reported a further reduction in malaria prevalence to 0.2% in children under the age of five years with considerable variation between the two islands (0.4% in Unguja and 0.0% in Pemba-TMIS 2017).

The annual malaria incidence has decreased from 5.3 in 2020/2021 to 2.7 in 2021/2022 per 1000 population. A total 4,849 of malaria cases were reported from both health facilities and community in this year, of which 261 were admitted in different hospitals Unguja and Pemba with total of 6 malaria deaths were reported. Magharibi 'B' reported highest number of malaria cases, followed by Mjini in Unguja while, Mkoani district have recorded the lowest number of cases of all districts.

The overall annual blood examination rate (ABER) was 30% slightly increased compared to the 27.15% of 2020/2021, of which 40% reached in Unguja and 14% in Pemba. However, malaria positivity rate decreased from 1.5 in 2020/2021 to 0.3 of the current reporting period. The below charts highlight the malaria burden for the past three years period in Zanzibar.

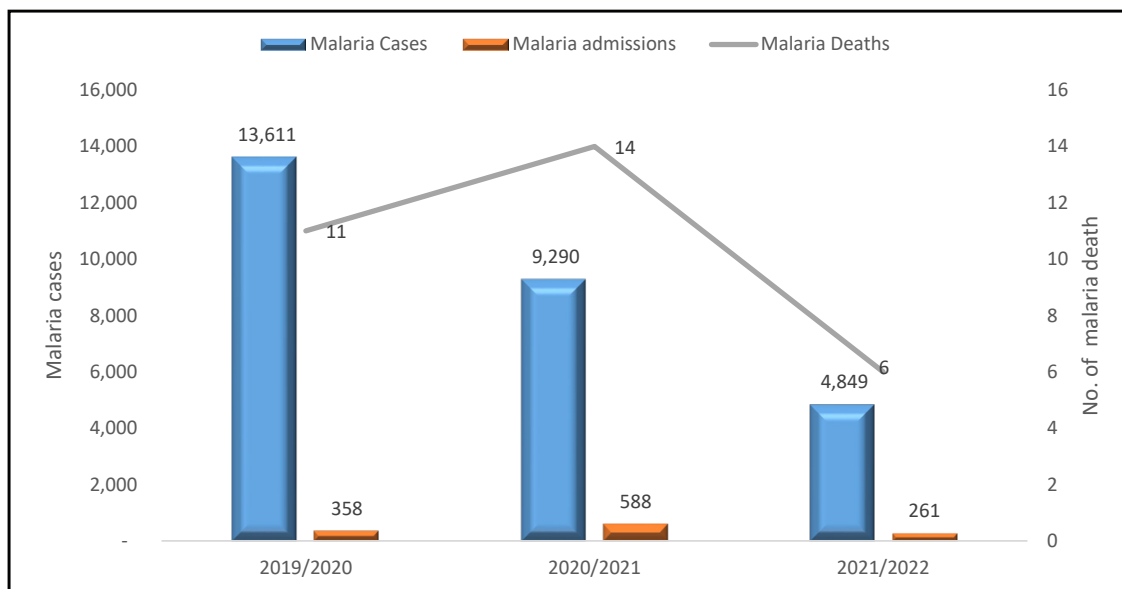


Figure 1: Number of malaria case, admission and death 2019/2020 - 2021/2022

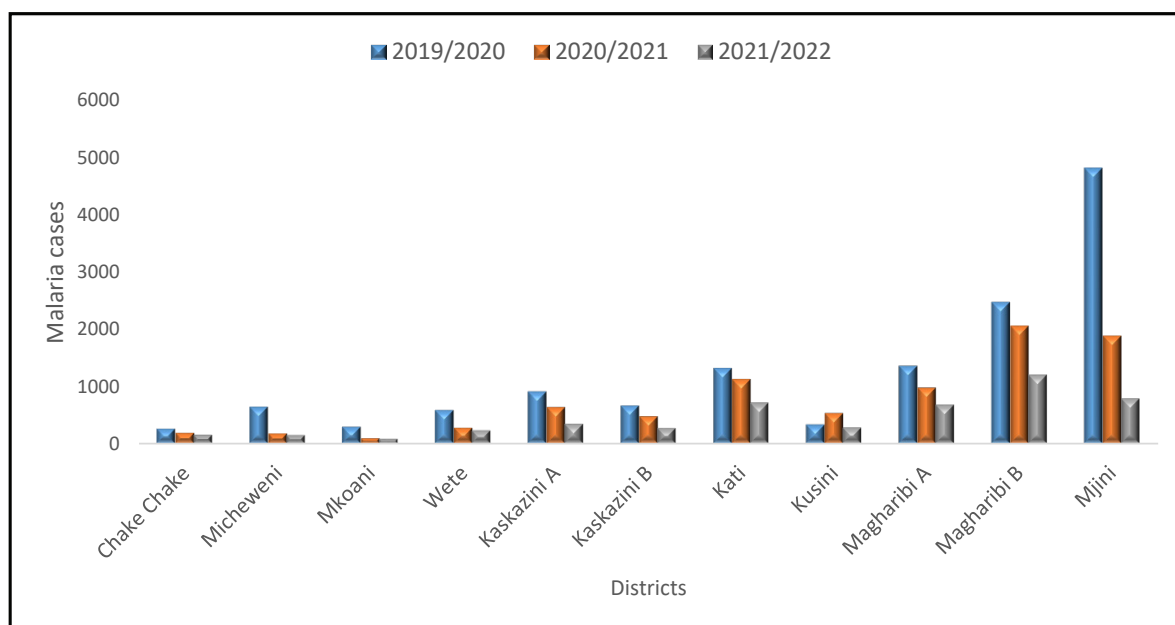


Figure 2: Number of cases by districts for 2019/2020 - 2021/2022



Zanzibar Malaria Vector

Surveillance data from routine malaria entomological surveillance from 10 ten sentinel sites indicates that *An. arabiensis* is still a predominant malaria vector species for more than 93%, in both Islands Unguja and Pemba; mostly of them exhibiting exophagic behavior. These outdoor tendencies exhibited by *An. arabiensis* may be occurred as a response to IRS and LLINs coverage leading to genetic changes of the vector species. Other Anopheles species identified in these surveillances include *An. merus* (3.5%), *An. gambiae* (0.37%), *An. lesoni* (0.1%) and *An. revulorum* 0.1%).

Although malaria transmission is at very low levels, highest peaks of transmission are always corresponds with periods of high rainy seasons with high mosquito abundance in the months of March to June and October to December every year.

Trend of Key Performance Indicators

For more than a decade now, reports on malaria implementation has shown good performance in the reduction of malaria incidence and other performance indicators. However more concerted efforts are required to fight against imported malaria case, strengthening multispectral engagement in fighting the disease and harnessing integrated larva source management as key intervention for sustainable malaria elimination in the country. The table below outlines performance of some of key malaria monitoring indicators in this year as compared with year 2020/21.

Table 1: Outlines Performance of Some Of Key Malaria Monitoring Indicators in this Year as Compared with Year 2020/21.

TREND OF KEY PERFORMANCE INDICATORS				
Indicator	2020/21		2021/22	
	Target	Achieved	Target	Achieved
Reported malaria confirmed cases	2,909	9,290	2,549	4,849
Annual parasite incidence: Confirmed malaria cases (microscopy or RDT): rate per 1000 persons per year (Elimination settings)	1.8%	5.3	1.6%	2.9
Malaria test positivity rate	0.90%	1.5 %	0.0	0.3%
Annual blood examination rate: per 100 population per year	46%	27.15%	0	30%
Proportion of patients with confirmed malaria who received first- line antimalarial treatment according to the national policy	100%	100%	100%	100%
Inpatient malaria deaths per year: rate per 100,000 persons per year	0.141	0.00008	0.12	0.00006
Percentage of confirmed cases fully investigated and classified	96%	88%	100%	92%



Proportion of suspected malaria cases that received a parasitological test (microscopy or mRDT) at public health facilities	100%	100%	100.00 %	96.80%
Proportion of suspected malaria cases that received a parasitological test at private health facilities	100%	100%	100%	97.70%
Percentage of health facility timely submitted monthly reports per the national guidelines	94%	50%	97%	92%
Proportion of health facilities without stock-outs of key commodities during the reporting period	100%	100%	100%	100%
Number of structures sprayed	30,000	46,316	15,000	21,470
Proportion of targeted households sprayed by IRS within the last 12 months	96%	100%	98%	100%
Number of LLINs distributed for use to the community	339,000	342,000	340,000	712,872
Number of long-lasting insecticidal nets distributed to targeted risk groups through continuous distribution	0	0	0	224,019
Proportion of population that slept under an LLIN in the previous night	88%	100%	85	100%
Proportion of households with at least one LLIN for every two people	100%	100%	100%	100%
Percentage of active malaria foci fully investigated and classified	100%	96%	100%	100%
Percentage of potential larva habitats in active foci in which larva source management implemented	87%	100%	97%	100%



CHAPTER ONE

PROGRAM MANAGEMENT AND COORDINATION

Zanzibar Malaria Elimination Program is guided by the National Strategic Plan ending 2023. At program level, there is a good platform that coordinates overall implementation of the planned activities in order to achieve the set goals and benefits. Comprehensive surveillance, monitoring and evaluation is the nuclear of the program that enable the authority to oversee and provides guidance to various thematic units in implementing the planned activities. In this reporting period, and in line with the current strategic plan, the programme managed to implement 85 out 88 planned activities with an average performance of 97.2%. Detailed information of implemented activities are found under each thematic area.

ZAMEP is continuing to receive remarkable support from the Revolutionary Government of Zanzibar and development partners mainly the GFATM and USAID through President Malaria Initiatives as stipulated in the section of malaria financing. Resource mobilization and utilization considering value for money to support efforts for malaria elimination in Zanzibar has been proven to be a major factor in achieving the program targets.

1.1 Strengthening Functional Partnership for Malaria Elimination

Zanzibar Malaria Elimination Program is still working with various development partners who provide financial and technical support for key malaria elimination interventions. The Zanzibar Malaria Elimination Advisory Committee (ZAMEAC) stands as a main independent board of advisors with members from wide range of malaria experts around the globe. This includes bilateral and multilateral partners, private sectors, research and academic institutions. The board usually meets once or twice a year to provide technically advice to ZAMEP in addressing operational hiccups and challenges towards elimination. For the past 3 years, all the board meetings have been conducted virtually or hybrid due to presence of covid-19 pandemic. The 4th ZAMEAC meeting was conducted in March 2022.

1.2 Program Procurement Management System

In line with Public Procurement and Disposal of Public Authority the Program's Procurement Unit ensures the application of fair, competitive, transparent, non-discriminatory and value for money procurement standards and practices. In the era of diminishing resources and increased demand for accountability and transparency in



government, the “stakeholders / shareholders” of the public sector are demanding more effective and efficient use of public resources.

Furthermore, ZAMEP is using standard operating guidelines for procurement of the necessary commodities as outlined in the annual procurement plans. Other relevant working documents are also used to assist program operation as planned for the year 2022/2023. Implementation of these plans are regularly monitored using specific tools with clear monitoring indicators per desired targets

1.2.1 Local Procurements

In this reporting period, a total of 423 procurements activities was planned according to annual procurement plan 2021/2022 costing a total of 595,402.77 US\$. Among those 196,507.61 USD went for 3 tenders under the Ministerial Tender Board since the amount exceeded the threshold. A total of 383,181.49 US\$ was planned for in-house procurement. In this case a total of 420 quotations was implemented, of which a total 1,260 vendors were invited for selection process and 421 suppliers was selected to supply various goods and services in Unguja and Pemba for both malaria and C19RM activities. The program is working closely with Ministerial Tender Board for approval of all item.

1.2.2 Pool Procurement Mechanism

ZAMEP adheres with development partner’s regulations in procuring health products for instance use of PPM. In this fiscal period a total of 2,031,991US\$ were allocated for procurement of health products whereby 1,086,573.46 US\$ were placed for ZAMEP and 945,417.37 were arranged for C19RM, Overall, number of orders placed through WAMBO was 10 of which 6 of them delivered and in use.

Table 2 : Summary of Fund and its Allocation for Procurement 2021/22

	TZS	USD
GE, C19RM, USAID and RGoZ	1,375,172,003.08	595,402.77
ZAMEP IN-HOUSE	885,015,128	386,181.49
MINISTERIAL TENDER BOARD	453,863,808.30	196,507.61
PPM ZAMEP & C19RM	4,693,188,013.15	2,031,991

1.3 Capacity Building

Based on the annual plan of action 2021/2022, ZAMEP planned various training for health care providers and community members as way of improving their expertise in moving forward the elimination agenda. A total of 1,718 health care providers have



been trained in various malaria interventions while at community level 287 influential persons were oriented on malaria strategies and interventions for implementation and reporting. Detailed information on various training conducted are available in the specific thematic areas.

1.4 Operational Research

In this reporting period, ZAMEP in collaboration with other institutions managed to conduct the following two research to contribute to planning and implementation of relevant strategies for malaria elimination

1.4.1 Zanzibar Malaria Importation Study

This is an observational research study jointly conducted between ZAMEP and MUHAS under the supported by University of North Carolina. The main objective of the study is to better understand the transmission of malaria from Mainland Tanzania to Zanzibar, and how it spreads within Zanzibar. This is an ongoing study with two years of sample collections followed by 2 years of data analysis ending at 2024.

The study intends to capture more than 90% of the malaria cases from 60 to 100 participating clinics in Zanzibar and twice of that number from 25 clinics in the mainland Tanzania that represent the top travel destinations. Data analysis will be done through genomics analysis to estimate the proportion of imported cases and extent of local transmission from index cases and finally mapping out the main malaria hotspots in Zanzibar.

1.4.2 Malaria Behavioral Survey

The main objective of this study was to provide better understanding of the socio-demographic and ideational characteristics associated with malaria-related behaviors in Zanzibar. The survey also aimed at determining appropriate focus of programmatic activities designed to improve malaria-related ideational and behavioral outcomes. It was conducted in the month of June to July 2021 and completed with data analysis and finding in December 2021. Summary findings of the study is attached with this report.

1.5 Key Programme Management Milestones

- i. Standard operating procedures for all components of surveillance have been prepared, field tested and are in use
- ii. There is a national reference laboratory for microscopy, with a slide bank and implementation of external quality assurance



- iii. An independent national malaria elimination advisory committee has been set up and regular meetings conducted
- iv. The national malaria elimination plan has been approved and endorsed by the Minister of Health
- v. The guidelines for inter-sectoral and multisectoral collaboration for malaria elimination has been completed and sent MoH higher authority for ministerial endorsement.



CHAPTER TWO

MALARIA CASE MANAGEMENT

2.1 Malaria Diagnostic Services

Prompt malaria diagnosis either by microscopy or rapid diagnostic tests is recommended by WHO and ZAMEP SP 2018-2023 for all patients with suspected malaria before treatment provision.

According to the current ZAMEP Strategic Plan the overall objective is to ensure quality assured diagnosis and appropriate case management in all health facilities and community level to 100% by 2023.

2.1.1 Implementation Status

In maintaining early case detection and confirmation by microscope and or mRDTs ZAMEP has the responsibility of ensuring malaria diagnosis commodities are available throughout a year. Storage and distribution of malaria commodities to all public health facilities is under the umbrella of Central Medical Store. The GFATM procured **13,267** of 25 test kits and OMDM via RTI using PMI funds procured several items for malaria microscope and monomolecular commodities. As a result, the annual blood examination rate increased compared to 2020- 2021 and 2021 – 2022 from 27% to 30% respectively. **(520,121)** tested from health facilities and community with the projected population of **1,717, 608**. No stock out of diagnostic commodities reported.

2.1.2 Expansion of malaria diagnosis quality assurance quality control tool at public and private health facilities.

ZAMEP conducted several activities to inform this strategy with the aim of ensuring malaria diagnosis results providing timely quality, reliable and accurate results. The following activities were conducted.

2.1.3 Capacity Building

In FY 2021-2022 a total of 340 (100%) were trained compared to 380 (85%) of participants in 2020-2021. These include 100 medical laboratory technicians trained on malaria microscopy quality assurance, 140 health care workers trained on malaria rapid tests quality management and 100 HCWs trained on preparation of thick and thin blood smear in hospital and public health facilities.

It was anticipated that the trainings conducted shall increase competencies of the medical laboratory technicians and other medical professionals in providing quality malaria results as per the required laboratory standards. Prior to training sessions, a pre



and later on post tests were conducted. The overall results showed that participants' knowledge increased from 35% to 85% respectively. Requirement for COVID-19 preventive measures were observed during the training

In this reporting period a total of **269 (mRDTs – 173 and microscope – 96)** health facilities were supervised on mRDT quality management and malaria microscope quality assurance quality control (mMQAQC) in public and private health facilities Unguja and Pemba Islands. Of which, 48,651 microscope slides were examined and confirmed positive were **663** and mRDTs tested was **339,976** and quality control was **>90%**; No invalid devised documented.

During malaria microscope quality assurance, the main categories were examined was detection, quantification and speciation. The results showed that; sensitivity was **96.4%** and specificity was **99.9%** after agreement with 1st and 2nd reader, the four districts revealed low performance sensitivity these are Magharibi "A", Mjini, Chakechake and Magharibi "B" were **88%, 90%, 93% and 94%** respectively while specificity was **97%** Kaskazini "A" districts. In parasite quantification the analysis showed that the minimum and maximum score of parasites density was 40 – 500p/μl [p- 0.005] respectively while on speciation *Plasmodium falciparum* remain as predominant species for proportion of 92.8% (615) followed by *Plasmodium malariae* 3.3% (22), mixed infection of *Plasmodium falciparum* and *Plasmodium malariae* was 3.3% (22) and *Plasmodium ovale* was 0.6% (4). No *Plasmodium vivax* detected from reporting period

Collection of blood smear slides from mRDT in 52 health facilities from Mjini, Magharibi A, Magharibi B and Micheweni districts. A total of 311 slides collected. Of which, the speciation's revealed as following; Pf 83% (258), mixed infection of Pf + Pm 8.7% (27), Pm 5.5 (17), Negative 2.3% (7) and auto- fixation 0.6% (2). All slides were examined with speciate and quantified by blood per microliters (μl) of parasites.

Proficiency Test (PT) was conducted in 64/ 96 (67%) HF's involved in mMQAQC provided with 10 blinded slides i.e. (4 -detection, 3- speciation and 3- quantification) with aim is to evaluating the performance of individual laboratories and laboratory technicians' performance for malaria test. This PT was conducted as per WHO guidance. The following graph shows the overall score in each round.

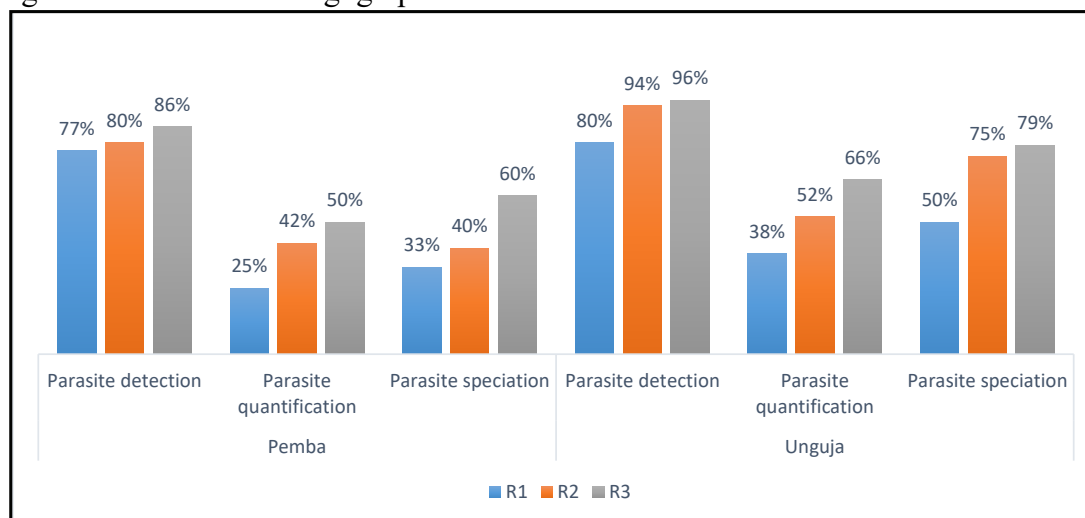


Figure 3: Proficiency Results in Zanzibar July 2021 – June 2022

Collection of blood smear slides from mRDT in 52 health facilities from Mjini, Magharibi A, Magharibi B and Micheweni districts. A total of 311 slides were collected of which, the speciation's revealed as follows; *Pf* 83% (258), mixed infection of *Pf* + *Pm* 8.7% (27), *Pm* 5.5 (17), Negative 2.3% (7) and auto- fixation 0.6% (2). All slides were examined with speciate and quantified by blood per microliters (μ l) of parasites.

2.2 Malaria Treatment

Malaria case management refers to the importance of early diagnosis and prompt effective treatment which remains a vital component of malaria control and elimination strategies.

Treatment of malaria using ACT's plus SLD primaquine and antimalarial for severe malaria in public and private health facilities.

2.2.1 Antimalarial Supply Chain System.

ZAMEP in collaboration with CMS continued to ensure un-interrupted supply and distribution of antimalarial in all public health facilities in Unguja and Pemba. The Government continued to support the procurement of antimalarial where no substantial shortage of antimalarials was reported except for perceived stock out of second line antimalarial ALU in 95 out of 110 (86%) public health facilities and missing of first line drugs for infants (ASAQ 25mg+67.5mg) in almost 78% (85) out of 110

public facilities. However, these stock outs did not affect malaria case management as most of the cases were successfully treated with first line regimens and no malaria cases reported among young infants.

2.2.2 Treatment of confirmed malaria cases with ACT plus SLD primaquine in both public and private health facilities

Training of health care providers on management of uncomplicated and severe malaria was conducted. The training involved 200 participants, (125 Unguja, 75 Pemba) overall achievement was 100%. Target was to train 200 participants from Unguja and Pemba.

2.2.3 Feedback Meeting on Death Audit, Death Audit Data Entry and Refresher Training

During this period four death audit meetings and data entry as well and one refresher training were conducted. A total of 61 participants were trained on death audit. (34 Unguja and 27 from Pemba)



Figure 4: Health Care Providers Participated in Death Audit Meeting

The purpose of death audit meeting was to discuss and set possible solutions for all challenges experienced in order to reduce the risk of similar death in the future. More importantly, is to share experience among the meeting participants to make assessment of all patients who died in this reporting period.

A total of audited death was 690 out of 4,079 (DHIS2; 2021/2022) total deaths from 9 public hospitals and 2 private hospitals, 3,497 Unguja, 582 from Pemba. During this period, 6 malaria deaths were reported and audited. It was noted that 66% of death is associated with delay arrival from the community and 34% was attributed to improper patient management at hospital level which calls for strict monitoring and on job technical support. The leading causes of death were complication of hypertension 50%, septicaemia 20%, road traffic injuries 17%, 12.2 other conditions and malaria 0.8%.

2.2.4 Supportive Supervision & Monitoring

Follow up supportive supervision in Unguja and Pemba was conducted in 180 health facilities (103 Unguja, and 77 Pemba) The target was 245.

The quality of services has improved such that the general performance was 73% compared with the base line of 64% for the year 2020/2021. During baseline it was noted that 44% of staff from public facilities were not aware on management of severe malaria and private facilities were 60%. As a result of intensive follow up, 75% of public facilities increased their knowledge on management of severe malaria and 52% for private sector.



Figure 5: Supervisor Collect Data at Health Facility During Field Visit

2.2.5 Dialogue Meeting

This activity was conducted biannually. Dialogue meeting involved 110 health care providers (60 Unguja, 50 Pemba). The aim was to discuss various matters on the provision of malaria related services including availability of malaria commodities, patients care and prompt treatment in order to reduce malaria morbidity and mortality.

2.2.6 Feedback Meeting.

Three meetings were conducted, involving hospital management, DMOs, DMSO, DP, LM, CMS to discuss logistics and supply of malaria commodities. The purpose was to identify districts with shortage of malaria commodities such as ACT, primaquine, injection artesunate and mRDT. Meeting members also discussed operational challenges related to overall malaria case management to health workers, following the recommendation redistribution of medicine were done regular through district pharmacists.



CHAPTER THREE

INTEGRATED MALARIA VECTOR CONTROL

The general objective of malaria vector control is to reduce the intensity of local malaria transmission by reducing the mosquito longevity, density and human-vector contact. The key vector control interventions in Zanzibar are Indoor residual spray, Long Lasting Insecticidal treated Nets and Larval Source Management. These main vector control interventions are reinforced by 3 malaria vector control activities namely surveillance at sentinel sites and hotspot areas, insecticide resistance monitoring and wall bioassays. The following 25 out of 26 major activities were implemented in this period with general performance of 97%:

3.1 Entomological Monitoring in 10 Established Sentinel Sites

Monitoring of entomological characteristics were done in all 10 established sentinel sites Unguja and Pemba (6 in Unguja, 4 in Pemba). On the other hand, the program through Vector control section continues to assess the impact of IRS and LLINs regarding the key parameter's potential for malaria transmission.

3.2 Distribution of Malaria Vector Species in Zanzibar 2021 - 2022

In 2021 – 2022 plans, the ZAMEP managed to analyze 1,872/2,890 samples of mosquitoes for mosquito species composition. Of the samples analyzed, 93% were *An. arabiensis* varying in districts and sentinel sites as shown in the table below:

Table 3: Malaria Vector Species in Zanzibar 2021 - 2022

Location	District	Sentinel sites	# of vector collected	# of vector tested	% <i>An. arabiensis</i>	% <i>An. gambiae s.s</i>	% <i>An. merus</i>	% <i>An. lessoni</i>	% <i>An. revulorum</i>
Pemba	Micheweni	Tumbe	353	310	66.9(236)	0	6.4(20)	0.96(3)	0.96 (3)
Pemba	Wete	Bopwe	810	591	100(591)	0	0	0	0
Pemba	Chake – Chake	Uwandani	148	99	100(99)	0	0	0	0
Pemba	Mkoani	Wambaa	129	18	100(18)	0	0	0	0
Unguja	Magharibi	Mwera	609	310	99(308)	0	0.96(3)	0	0
Unguja	Kati	Cheju	694	471	99.5(469)	0	0.5(2)	0	0



Unguja	Kaskazini A	Donge	28	21	67(14)	9.5(2)	23.8(5)	0	0
Unguja	Kaskazini B	Bumbwini	71	12	67(8)	25(3)	8(1)	0	0
Unguja	Kusini	Muyuni	0	0	0	0	0	0	0
Unguja	Mjini	Mjini Mkongwe	48	40	7.5(7.5)	5(2)	87.5(35)	0	0
Total			2890	1872	93(1,746)	0.37 (7)	3.5(66)	0.16(3)	0.16(3)

3.3 Vector Biting Behaviour and Infection Rate

The *Anopheles arabiensis* are currently found to have exophagic characteristics, of which 74.75% (83.5% in Pemba and 66% in Unguja), indicating that malaria transmission occurs when people are still active, and high age groups have a chance to contract the infection due to their activities. This suggests the importance of understanding exactly where vectors and humans interact. The average vector infection rate was 0.54%, varying in sentinel sites, being higher in Unguja (0.68%) than in Pemba (0.4%). All infected vectors were morphologically identified as *An. gambiae* s.l, possibly *An. arabiensis* and *An. merus*. The mosquito infection rate is lower compared to last year by 0.26 percentage points, indicating the effectiveness of vector control interventions. This finding suggests a decrease in indigenous malaria cases.

3.4 Insecticide Resistance Monitoring, Intensity and Mechanism in Malaria Vectors

ZAMEP conducts insecticide resistance monitoring to malaria vectors in April - June every year when the mosquito population density is on the higher side. The objective is to understand the susceptibility of malaria vectors to insecticides commonly used for public health. In 2022, Pyrethroid resistance (Deltamethrin, Permethrin and Alpha cypermethrin) to *Anopheles gambiae* s.l. was still stable in all ten (10) sites of Unguja and Pemba. However, they were full susceptible to Clothianidin, Pirimiphos methyl and bendiocarb.

Table 4: Susceptibility Test Results of Malaria Vectors to Various Insecticides 2021 – 2022

Location	District	Sentinel Site	Percentage mortality					
			Permethrin	Deltamethrin	alpha-cypermethrin	P-methyl	Bendiocarb	Clothianidin
Unguja	Magharibi	Mwera	86	93	76	100	100	100
Unguja	Kaskazini A	Donge	76	86	76	100	100	100
Unguja	Kati	Cheju	78	80	85	100	100	100
Unguja	Kusini	Muyuni	84	92	73	100	100	100
Unguja	Kaskazini B	Bumbwini	82	94	85	100	100	100
Unguja	Mjini	Stonetown	83	94	77	100	100	100
Pemba	Wete	Bopwe	40	30	27	100	100	100
Pemba	Micheweni	Tumbe	65	68	66	100	100	100
Pemba	Mkoani	Wamba	60	71	65	100	100	100
Pemba	Chake Chake	Uwanda	22	15	21	100	100	100

Key <90% = Resistance, 90% - 97% = Possible Resistance, > 98% = Susceptible

The strength of pyrethroid resistance indicates *An. gambiae* s.l. had high resistance to X5 and X10 diagnostic doses of alpha-cypermethrin, Permethrin and Deltamethrin. The intensity of pyrethroid resistance was not homogeneous at sentinel sites and ranges from low to moderate and high. Pyrethroid resistance is high in Zanzibar. However, we are confident that there is no disease control failure within the community as PBO nets provide additional prevention in malaria protection by fighting metabolic-resistant mosquitoes.

3.5 Insecticide decaying rate in treated surfaces following Focal IRS using Fludora fusion

In early August 2021, ZAMEP conducted a focal IRS as a response using fludora fusion in two selected villages Pemba and Unguja. This was done following abnormally increase in local malaria cases where. Over 98% of the targeted structures were covered in this operation. Fludora fusion is a combination of Clothianidin and Deltamethrin used at a dosage of 300mg AI/m². Clothianidin is a slow-acting insecticide formulation. Hence, the usual WHO protocol for cone bioassays allows mosquito mortality observation after every 24 hours for one week. Deltamethrin increases the knock down to the vectors.

The results indicate that the Fludora fusion is a promising insecticide for malaria control due to its quick knock-down effects, high toxicity to the vectors and long residual effect in treated walls. The insecticide is potential for future use in ZAMEP. Findings are found in the graphs below:

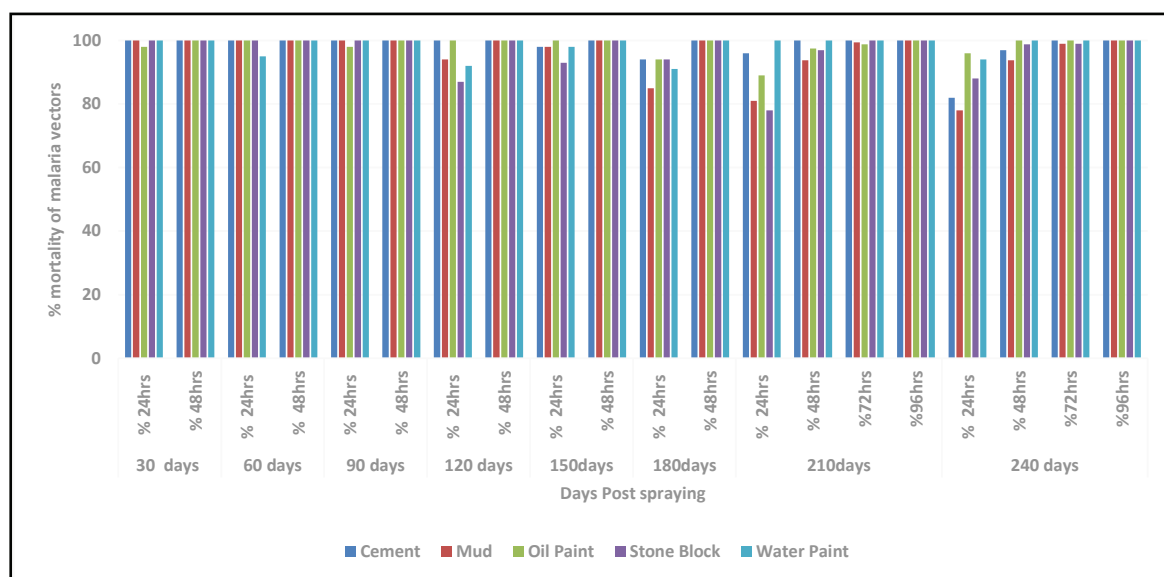


Figure 6 Fludora fusion efficacy residue from one to eight months in two selected villages in Pemba, 2021 /2022

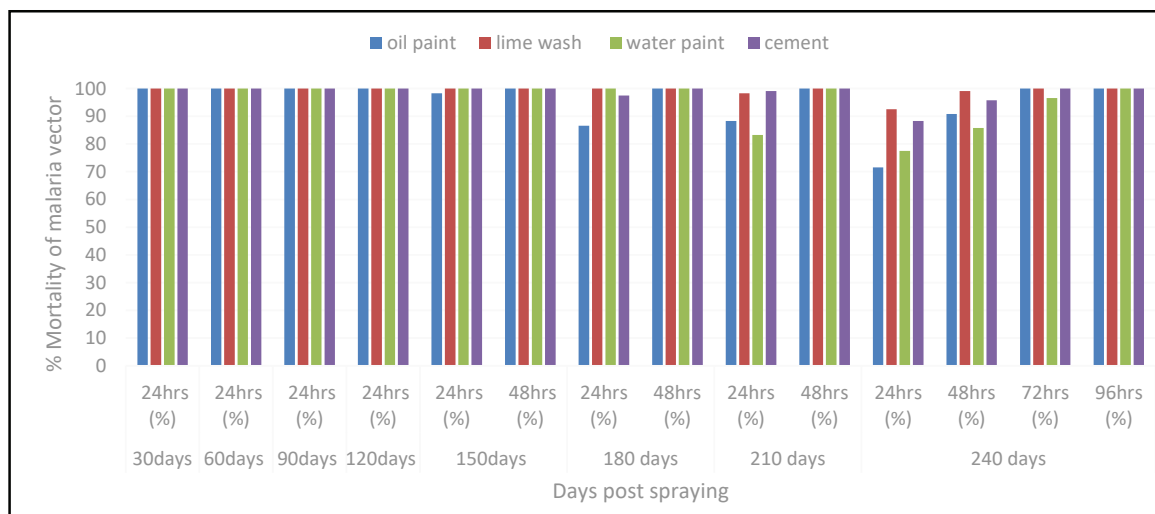


Figure 7 Fludora fusion efficacy residue from one to eight months in Unguja 2021 /2022

3.6 Foci investigation and response

The main objective of the activity was to investigate entomological risk factors and take appropriate response when necessary. The variable of interest includes vector and mosquito habitats characterization. ZAMEP has recorded 396 Foci (131 in Pemba and 265 in Unguja).

Of these, 46.4% (n = 184) are active, 49% (n = 195) are residual non-active, and 4 % (n = 17%) are cleared. The active foci vary in the district and island.

In period of 2021/2022, ZAMEP planned to investigate 100 active malaria foci. Of these, total of 99 (99%) have been investigated in this year compared to 72 of last year. Of the investigated foci, ZAMEP conducted reactive focal IRS using clothianidin in 64% of them (n = 63) and cover 98% of targeted structures (n = 21,470), protecting 98% of the population (n = 130,016).



Table 5: Foci investigation and Reactive IRS response per district 2021- 2022

District	Target of foci investigation	Proportion of foci investigated	% of active foci protected by IRS	% Structures sprayed	% population protected by IRS
Wete	20	100 (n = 20)	100 (n = 20)	97.6 (n = 3,276)	98.2 (n = 18,365)
Mkoani	6	100 (n = 6)	100 (n = 5)	99.8 (n = 1080)	99.9 (n = 5,907)
Chake Chake	5	100 (n = 5)	80 (n = 4)	98.6 (n = 955)	98.9 (n = 5,596)
Micheweni	16	100 (n = 16)	94 (n = 15)	97.6 (n = 3,737)	97.4 (n = 20,760)
Mjini	11	100 (n = 11)	9 (n = 1)	100 (n = 47)	100 (n = 358)
Magharibi A	5	100 (n = 5)	60 (n = 3)	98 (n = 963)	95 (n = 6,356)
Magharibi B	15	100 (n = 15)	47 (n = 7)	95 (n = 9,281)	96 (n = 66,586)
Kati	11	100 (n = 11)	55 (n = 6)	97 (n = 1,078)	96 (n = 6,075)
Kusini	3	100 (n = 3)	67 (n = 2)	98 (n = 1,039)	97 (n = 5,841)
Kask A	6	60 (n = 5)	40 (n = 2)	100 (n = 14)	100 (n = 79)
Kask B	2	100 (n = 2)	0	-	-
Total	100	99 (99)	64 (n = 63)	98 (n = 21,470)	98 (n = 130,016)

In foci entomological characterization, the team collected 789 *Anopheles* vectors. Five hundred and thirty three (533) were molecularly analysed into species level. *Anopheles arabiensis* was a predominant at 88% with highly exophagic behaviour (64%). *Anopheles gambiae* s.s was reported in less than 1% in both Unguja and Pemba. The species is known to be excellent and competent in malaria transmission due to susceptibility to infection, longevity and anthropophagic behaviour. It is an alert message to ZAMEP that the competent vector is within our environment, and vigilant monitoring /surveillance is needed as the species can reverse the programme into a control error. High out door biting behaviour of the vector is reported in Pemba (72%) than Unguja (55%). The vector infection rate in foci was zero in 700 samples screened for sporozoite rate.



3.7 Physical integrity and efficacy of Olyset plus

ZAMEP conducted a mass net replacement campaign last July 2021 by distributing 730,950 nets (286,840 in Pemba and 444,110 in Unguja) nets to cover the population of almost 1.6 million in a maximum of four pieces per household. The main objective was to conduct a prospective follow up of the campaign nets (Olyset Plus) regarding attrition/retention, and insecticide efficacy against susceptible *Anopheles gambiae* s.s R70 using a WHO bioassay cone. In Pemba, 71% (n =137) and in Unguja, 36% (n =166) of the nets were still available in sampled households. Many of the lost LLINs have been reported mostly to be given to relatives to students and disposed off due to damage.

The average results of bioassay tests against susceptible *Anopheles gambiae* ss from ZAMEP insectary showed the mean mortality of 45.6% in Pemba and in 75% in Unguja. However, the the recommended WHO performance is 80% .The findings call for a close investigation to monitor the effectiveness of PBO nets against malaria vectors. The three-year life span for LLINs is doubtful and can be considered a stereotype concept.

3.8 Larva Source Management (larviciding) in active foci

The increased outdoor biting of malaria vectors forced ZAMEP to introduce larviciding to complement the core interventions (IRS and LLINs) to reduce receptivity within the community.

In 2021/2022, ZAMEP recorded 290 (221 in Pemba and 69 in Unguja) potential water bodies for mosquito breeding that can survive for at least four months per year in **88** Shehias. The breeding sites include ponds, streams, marshes, mangrove swamps, rice fields, caves and unused hand-dug wells. Many water bodies were close to residential houses, less than one kilometre within the malaria vector flying range. The number of breeding sites varied between districts and shehia. However, many of them had natural vegetation ranging from short, long, mixed and float, with *Anopheles* and *culex* larvae of different stages.

In this reporting period, ZAMEP conducted reactive focal larviciding using bti in 63 active foci, whereby 188 mosquito breeding sites were treated in one to two rounds (88 in Pemba and 100 in Unguja.). WHO recommend a minimum of six to eight bti larviciding rounds to have an entomological and epidemiological impact on disease transmission, accompanied with effective monitoring. However, due to shortage of resources such as funding and other logistics did not happen in Zanzibar, and therefore it is difficult to describe its contribution in disease case reduction.



3.9 TOT training for IRS rapid response

ZAMEP, in collaboration with ABT/Vector link, conducted a TOT training on indoor residual spraying comprising various IRS cadres, including supervisors, logistics (store men), monitoring and evaluation and environmental compliances. The training involved 26 participants from ZAMEP and all districts of Unguja and Pemba. The main objective was to prepare a reactive focal IRS team regarding planning, implementation and monitoring of IRS rapid response.

CHAPTER FOUR

SURVEILLANCE MONITORING AND EVALUATION

Surveillance, monitoring and evaluation is one among core intervention earmarked for malaria elimination in Zanzibar. Main objective is to reinforce surveillance of malaria cases by actively investigate and classify all confirmed cases with their epidemiological related information for structuring effective and timely interventions. The following main activities were implemented in this period to strengthen surveillance activities:

4.1 Malaria stratification

The malaria stratification is crucial for Zanzibar to meet its elimination target: it enables ZAMEP to target interventions more effectively to those areas that report the highest burden of malaria (and thus are likely to be areas where the population is at greatest risk of malaria), thereby allowing for more effective programmatic resource allocations and maximizing the interventions' impact.

The methodology used to stratify the shehia was based on key determinants of malaria transmission. For instance, malaria burden (weekly case load and annual incidence) in the last 3 years, vulnerability in terms of population movement, and probability of finding new cases during index case investigation. These key determinants were given weights to stratify the malaria burden. The table below shows the result of stratification using data from 2017 to 2021.

Table 6: Number of shehias documented to each stratum between 2017–2019, 2020, 2021, and 2019–2021 data

Data	High diffuse transmission	Moderate focal transmission	Low focal transmission	Pre-elimination
2017–2019 (baseline)	Zanzibar=54 Pemba=6 Unguja=48	Zanzibar=97 Pemba=16 Unguja=81	Zanzibar=136 Pemba=52 Unguja=84	Zanzibar=100 Pemba=55 Unguja=45
2020	Zanzibar=97 Pemba=18 Unguja=79	Zanzibar=92 Pemba=10 Unguja=82	Zanzibar=85 Pemba=34 Unguja=51	Zanzibar=113 Pemba=67 Unguja=46
2021	Zanzibar=97 Pemba=5 Unguja=92	Zanzibar=75 Pemba=35 Unguja=40	Zanzibar=30 Pemba=4 Unguja=26	Zanzibar=185 Pemba=85 Unguja=100

4.2 Improve data quality assurance

In effort to improve data collection and reporting at all levels, regular supportive supervisions was conducted in 104 health facilities and 28 DMSOs. In this period, total of 4,677 (95%) malaria cases were investigated and reported by DMSOs before being verified by the ZAMEP team. Findings from these supervisions has shown that 95% of cases were documented and classified correctly in malaria case notification registers.

Reporting timeliness has improved from 89% to 95% and consistency of data from 95% to 98% compared to the last year. Moreover, regular feedback and data review meeting were conducted involving 346 health facilities staff and 28 DMSOs with the aim of improving malaria data quality in MEEDS/IDSR and MCN systems.



Figure 8: Data Review Meeting

4.3 Strengthening data usage and dissemination

Malaria surveillance data was disseminated to health facility workers, district authorities through feedback meetings, weekly summary reports, quarterly and annual reports. The aim was to provide updates on surveillance activities including malaria trends, (uncomplicated cases, admission & deaths) for programmatic decision.

4.4 Enhancing Case investigation and classification

Several trainings, and bi-monthly feedback meetings were conducted in this period to improve the capacity to DMSO's, health facility staffs and DRT's. A total of 349 health facilities Staff from Unguja and Pemba were attended these sessions.

In this period, a total of 4,849 malaria cases were detected and reported from both public and private health facilities. Out of these 4,395 (95%) of the cases were fully investigated up to household level. During the household investigation additional of 490 people were tested and (8%) malaria positive case was detected and treated accordingly. Out of investigated cases, 89 % were classified as imported, and 11% were classified as indigenous cases.

4.5 Empowerment of district Response Teams to conduct response activities

DRTs conducted 38 ACDs in different villages of shehias that reported an increase of malaria cases. The coverage was 87.8% whereby 22,103 people from the community were tested, 45 (0.2%) were detected with malaria parasites. In addition to ACD, vector control, and SBCC interventions were conducted aiming at reducing malaria transmission in those areas.



Figure 9: Malaria Case Verification



CHAPTER FIVE

SOCIAL BEHAVIOURAL CHANGE COMMUNICATION

SBCC is one of the main supportive strategies with an objective of enhancing advocacy, behavior, social communication and mobilization on malaria elimination intervention to general population. In this financial period, the program under the support of Global funds and USAID managed to conduct six activities with the aim of raising awareness and providing technical support to the district and community members on key malaria elimination interventions. Key implemented activities in this period include:

5.1 Harmonization of by-laws to district implementers

This activity was implemented twice in this reporting period involving a total of 120 participants (60 from Pemba) including district lawyers and district planning officers. The activity was done as advocacy meeting for district authority on the new public health regulations that have been approved since 2012 by the Zanzibar Central Government. As a result of these meetings, the by-laws are now in operation by the authority especially on the areas of Indoor Residual Spraying and proper-utilization of LLINs at community level.

5.2 Providing support to the district authorities to perform health education activities

This was an educative meeting which invited a total of 85 participants from Unguja and Pemba (40 Pemba). The objective of the meeting was to provide technical support to district health management teams targeted to district health educators who are mainly involved in conducting and providing health education promotion on malaria. Among the training contents were preparation of lesson plans, preparation of sequence of events and channel of communication that can be used to disseminate malaria information to general public.

5.3 Training of community members on malaria elimination strategies

The training of community owned resource persons was implemented with an intension of increasing community knowledge on malaria elimination strategies. The objective of the training was to orient community members including CHV on the package of key malaria elimination interventions to be able to provide support and boost malaria elimination movement at grass root level. A total of 182 community members from 33 Shehia with high number of malaria positive cases were invited. The outcome of the training is to strengthening existing community health committees and CHV work that are in line with malaria elimination interventions. The involvement of CHV to malaria



elimination activities has been insisted by the MOH in order to strengthen public health interventions including malaria elimination at Shehia level.

5.4 Mass Media communication

Mass media campaign is used to disseminate malaria elimination to general public. Live and recorded radio and television programs, production of billboards and broadcasting of radio spots messages are among the common mass media communication used by ZAMEP.

During this reporting period a total of 45 live radio programs and 35 television programs have been prepared and aired through different media available in Unguja and Pemba. Furthermore, a total of 3 billboards with different malaria elimination messages have been printed and placed to strategic areas for awareness raising, mainly at the ports of entry and main road junctions. Key messages covered in these programs include importance of early health seeking behavior, malaria prevention for travelers and importance of use of ITN for malaria prevention.

5.5 Malaria Behaviour Survey

In 2021 the Malaria Behaviour survey) was conducted in order to understand on necessary information regarding key malaria interventions to support malaria elimination movement. Some of the major findings of the survey are outlined in table below:

Table 7: Key Findings of the Malaria Behaviour Survey 2021

SN	Key interventions	Proportion 2021	Urban	Rural	Remarks
1.	Households owned at least one ITN	65	56.6	70.0	This finding obtained before MRC 2021
2	Knew about using bed nets (treated/untreated) to prevent malaria	91	90.2	91.7	
3	ITNs Utilization	85	90.0	80.0	
4	Awareness for IRS program	58	59.1	57.5	
5.	Awareness for larviciding	29	42	17`	
6.	Aware of malaria MDA programs	16	17.2	15.8	
7.	Aware of RCD programs	24	24.0	24.4	
8.	Identified ACT as the medicine to treat malaria	15	8	7	
9.	Know Malaria is Caused by Mosquito Bite	86	85.2	86.1	
10	Know at-least one major prevention measure	92.2	93.2	92.0	



5.6 School Health Program

The aim of this activity was to bring close malaria elimination messages and involve school pupils in malaria elimination at school level. The malaria clubs have a role of disseminating the malaria elimination messages to fellow school pupils and nearby community members. A total of 176 school teachers from 30 schools (15 schools from Pemba) were trained on the key malaria elimination strategies. The training was aiming at formulating malaria school clubs of which a total of 30 malaria school clubs have been formulated for the year 2021.

5.7 Interpersonal communication for malaria elimination

The strategy is involving community meetings and dialogue to sensitize community members on key malaria elimination interventions. This channel of communication provide great chance of interaction between facilitators and audiences of which questions are asked and answered online. For the year 2021 a total of 54 community meetings were conducted particularly to the Shehia that reported high number of malaria cases. This is similar to 11 sensitization meetings that have been conducted in 11 secondary schools from Urban West region.

Among the topics discussed were importance of sleeping under the ITNs continuously and cleaning of living environment for the purpose of abolishing mosquito breeding sites.

5.8 Monitoring and supervision of SBCC

This is an essential part of SBCC activities that assessing the progress of implemented planned activities in a quarterly basis. For the year 2021/2022 all 388 Shehia were visited as part of monitoring and supervision. Similarly, 180 health facilities from Unguja and Pemba were also visited. At Shehia level the supervisors assessed the functionality of the formulated community health committees, presence of community sensitization plan and IEC materials, these also assessed at health facility level.

Out of 388 visited Shehia a total of 102 (26%) Shehia did not have work plan and not met or worked as planned for more than eight months. However, a total of 180 visited health facilities identified to have at-least one malaria IEC material and having a lesson plan.



5.9 ITNs distribution

The program is using two distribution channels to ensure ITNs are available to targeted groups throughout a year. In so doing, ZAMEP implementing integrated distribution approach that involve Central Medical Store, HIMS, LMU and USAID Tanzania Vector Control Activities Project. Below are the main activities that were implemented to support ITNs distribution for the year 2021/2022.

5.9.1 Continuous net distribution scheme/channels

Through continuous net distribution scheme that targeting pregnant women and children at nine months during the measles and rubella1 vaccination, a total of 99,789 nets have been issued to those two major target groups as stipulated in the below table:

Table 8: Number of LLINs issued during MRC, 2021

Target groups	Annual target	Number of Net issued annual	Remarks
Pregnant women	40,000	52,077	The distribution exceeded the annual target following the cessation of community distribution channel whereby additional nets were issued
Children at the age of nine months	40,000	47,712	
Total		99,789	

The ITN services are available in 180 health facilities of Zanzibar. Among the documented challenge is slow community collection of nets at health facility level following the communities receiving net from Mass Replacement Campaign.

5.9.2 Mass Replacement Campaign

Mass Replacement Campaign was completed in 335 registered Shehia for the year 2021 (129 from Pemba). The objective of the Campaign was to increase LLINs ownership among community members leading to utilization to the general public.

In 2021, mass replacement campaign targeted a total of 746,420 LLINs to be distributed as per the registration at the household level. However, at the end of the distribution the Program managed to distribute a total of 712,872 LLINs (96%) Unguja and Pemba.



Table 9 : Number of nets issued in each district

District	Quantified/ Target	LLINs Issued	% of LLINs Issued
Mjini	56,065	49,491	88
Kaskazini A	82,147	80,502	98
Kati	58,642	57,389	98
Magharibi A	90,445	84,779	94
Kusini	26,467	25,116	95
Kaskazini B	53,799	50,523	94
Magharibi B	82,825	78,232	94
Wete	80,853	78,448	97
Mkoani	71,857	68,642	96
Chake	73,473	71,149	97
Micheweni	69,847	68,601	98
Total	746,420	712,872	96

5.9.3 Monitoring and supervision of ITNs

This has been conducted in each quarter to all 180 Health Facilities that providing services of net distribution at ANC level Unguja and Pemba. Among the challenges identified were lack of palates for storage of nets, delaying in requesting nets through R&R system and shortage of trained health workers to some health facilities especial from Pemba. All identified challenges were worked out jointly between ZAMEP and district health management teams.



CHAPTER SIX

RESILIENT AND SUSTAINABLE SYSTEMS FOR HEALTH

The Zanzibar Malaria Elimination Programs is working with different Ministry of Health units and institutions with the aim of strengthening the system and building the resilient and sustainable health system in the country. For that purpose, the ZAMEP worked with the Zanzibar Food and Drugs Agency, Health Promotion Unit, MoH Logistic Management Unit, Health Management Information System, and Emergency Operation Center in this reporting period

6.1 Zanzibar Food and Drugs Agency

ZFDA is a semi-autonomous agency established under section 3 (1) of Zanzibar Food, Drugs and Cosmetic Act No 2 of 2006 with the aim of protecting public health through improving food quality and its safety. This is done by regular inspection and control of food, cosmetics and medicine entered and circulating in the country.

In strengthening the quality of laboratory services and building capacity in quality assessment and the quality assurance of the drugs, Global Fund through ZAMEP supported ZFDA to procure the Gas Chromatography - Mass Spectrophotometer (GC-MS). The machine will be used to assess the quality of antimalarial drugs and other related commodities including bioanalysis of body fluids to detect presence of narcotics, barbiturates, alcohols, and drugs such as anticonvulsants, anesthetics, antihistamines, sedative hypnotics, and anti-epileptic drugs.

Presence of this machine will reduce the cost of transportation and testing of such items to Tanzania Mainland, hence facilitating timely results. Total of 12 ZFDA Analysts have been trained on operating procedures of this machine. The processes for installation and use of the machine are going on at final stage.

6.2 Health Promotion Unit (HPU)

Health Promotion Unit in collaboration with ZAMEP, managed to conduct several activities to strengthen and facilitate implementation of Community Health Strategy in line with Malaria Elimination.

The unit has developed several flipcharts (Bango Kitita) as a tool kit for Community Health Volunteer to facilitate community sensitization. Total of 495 flipcharts have been printed and distributed to CHVs in Mjini Magharibi region.



Figure 10: Participants during the development of CHV Flip Chart

The Unit also conducted community sensitization using community theater art and mobile vans aiming at improving knowledge and practices on malaria prevention and control. This was conducted in 3 strategic centers where a total of 75 Shehia have been reached.



Figure 11: Theater group demonstrate the proper use of LLIN to the community

Furthermore, CHV data monitoring tools have been developed and incorporated into Jamii ni Afya App to facilitate regular monitoring of CHV performance. The tools are on the process to be incorporated in DHIS2 platform.



6.3 Logistic Management Unit (LMU)

Logistic Management Unit is responsible for organizing, monitoring, and supporting all supply chain activities in all public health facilities in Zanzibar.

In this reporting period, two main activities were implemented to strengthen the public health supply chain performance. These include Assessment on rational use of antimalarial commodities and Procurement of computer and hardware for implementation of eLMIS facility edition.

6.3.1 Assessment on rational use of antimalarial commodities:

The assessment was conducted from January to June 2022 in two districts with high malaria incidences (Mjini Unguja and Micheweni Pemba). This is a short descriptive assessment that collected quantitative and qualitative data to describe the characteristics of the population on the practice among the prescribers and the level of dispensing practice of antimalarial medicines in public and private health facilities and community pharmacies

The assessment came up with the number of following main findings, here are some of them

- About 71.6% of the prescriptions adhered to National Malaria Diagnosis and Treatment Guideline while more than a quarter of the prescription didn't follow National treatment guidelines.
- Sixty-five (65%) of the study population obtained medicines from the health facilities' dispensary, and 33.8% obtained the medicines from OTC or neighbor and friends.
- About 46.5% of the clients who went to private retail pharmacies, purchase antimalarial medicines without having a formal written prescription.
- Only 42% out of 114 interviewed individuals who were treated for malaria, had evidence of confirmed malaria diagnosis.

Finally, the assessment team have concluded that, the problem of self-medication for malaria medicines seems to be still high within the population, hence more education and sensitization effort are needed among healthcare workers and entire community if sustained malaria elimination to be attained. Final report of this assessment have been shared within Ministry of Health, Local and International NGOs and development partners for their evidence based planning.



6.3.2 Procurement of computer and other hardware facilities for implementation of eLMIS - Facility Edition

The e-LMIS - facility edition has been established in 2019/20 in 3 district hospitals, Kivunge, Makunduchi, and Mkoani Pemba, with the aim of extending to all district and regional hospitals, Unguja and Pemba. During this reporting period, total of **34** computers and other hardware items were procured under the GF/ZAMEP support in the effort to extend and strengthen the system. These include 36 pcs of UPS, internet routers, UPT cables, antivirus, etc.

6.4 Health Management Information System (HMIS)

The main responsibility of HMIS Unit is to ensure timely availability of health and health related data and information from different sources within the Ministry of Health. Hence, HMIS is responsible for comprehensive data collection, storage, analysis and dissemination of information in different platforms of health information including MoH dashboards, annual health bulletins and other reporting media at all levels of health system.

During this reporting period, 93.8% of targeted health facilities were supervised for the purpose strengthening the data quality and management at all levels. The findings from supervisions shows that, 81.6% of the facilities had all required tools for collection, summary and management of data throughout a year. All (100%) of the visited HF's with ILLNs distribution services during the ANC visit managed to document all required information as directed in the respective register books. Moreover, reporting timeliness and completeness during time of supervision was 95% and 98% among all health facilities, respectively.

Other key findings and challenges observed during supervisions includes use of outdated registers in some health facilities, inadequate capacity in data collection and management among some/untrained healthcare workers and some counting errors during data collection. Despite of these challenges, most of the facilities managed to submit their report forms to the DHMT timely. Two feedback meetings after supervisions were conducted in all districts where, all findings and proposed solutions for the identified gaps were discussed for further improvement.



6.5 COVID-19 Response

Like all other countries around the globe, Tanzania and Zanzibar in particular have been hugely affected by COVID-19 pandemic with vast health and socio-economic impacts. The first case of the diseases in Zanzibar was diagnosed and reported by health authority on 18th March 2020. As of end of August 2022, the COVID-19 epidemiological situation report (SITREP) indicates that, total of 750 cases have been confirmed positive of the disease out of 2,761 suspected and tested individual in 68 selected health facilities, Unguja and Pemba. Total of 37 deaths from the disease have been reported in this period with CFR of 5%. On the other hand, total of 269,750 people have been fully vaccinate with Covid -19 vaccines making the total vaccine coverage of 36.2% of the targeted population.

In responding to the pandemic, the government has demonstrated a high-level leadership by taking swift actions to fight the ongoing Covid-19 pandemic, at the same time, tackling the potential risks associated with the pandemic and ultimately flattening the curve its transmission. The Zanzibar Public Health Emergency Operation Center (ZPHEOC) act as main hub for regular coordination of all preparedness and response activities during COVID-19 pandemic and other potential public health emergencies. The MoH also, worked hand in hand with the Disaster Risk Management Commission of the Second Vice President's Office in coordinating the COVID-19 pandemics mitigation activities.

Under the support of ZAMEP/GF, the ZPHEOC with other collaborating units managed to implement the following 6 six main activities in the effort to contain the pandemic and it effects to the public:

6.5.1 Quarterly supportive supervision and mentorship to all designated and other HCFs in Unguja and Pemba:

Main objective of these supervisions was to improving screening, triage, case identification, isolation and testing for COVID 19 suspect at HCF. Total of 51 (35 Unguja and 16 Pemba) selected Health Facility were visited. It has been emphasized that, to ensure effective management of the pandemic, all the screening, diagnosis and treatment of cases should follow available standards and protocols.

6.5.2 Refresher training on molecular biology using Biorad PCR machine:

This was two weeks activity conducted in Unguja and Pemba with main objective of increasing knowledge that will enable effective and efficient use of PCR machines with sufficient and competent expertise on Molecular technologies. Total of 47 laboratory technicians from Unguja (27) and Pemba (20) participated this training.



6.5.3 Refresher training to district surveillance officers:

Total of 231 facility supervisors and district surveillance district surveillance officers (116 Unguja and 115 Pemba) have been trained on early warning and response (EWAR), basic concept of Indicator-Based Surveillance (IBS) and use of One Health Approach in fitting covid-19 and other epidemic diseases.

6.5.4 Orientation training on interpersonal (IPV) and other form of Violence to district law enforcement officers

Total of 300 participant (195 Unguja and 105 Pemba) were oriented knowledge and practices on prevention and elimination of interpersonal violence in the community particularly during crises like covid-19. The training emphasized on strengthening collaboration and cooperation between victims and specific sectors which deal with violence in order to minimize psychological harm, death, and injuries resulted from violence.

6.5.5 Support to district surveillance officer on advanced incidence management system (IMS):

PHEOC also have been providing support to the districts on incidence management system (IMS) aiming at improving coordination on the response of any public health emergency at national and district levels. This has contributed a lot in strengthening capacity of the districts' teams to be able effectively utilize the available human and material resources in fighting public health emergency.



CHAPTER SEVEN

MALARIA FINANCING

Financial resources for malaria interventions within ZAMEP, is obtained from the Revolutionary Government of Zanzibar and development partners mainly GFATM and, PMI/USAID through RTI.

7.1 Basis of preparation

The financial annual report has been prepared under the historical cost basis except for motor vehicles and land that have been measured at fair value. The financial annual report is presented in US\$ unless otherwise stated.

7.2 Annual Report Compliance

The financial annual report of the Program has been prepared in accordance with Grant Agreement, ZAMEP Public Finance and Procurement Management Act (2016), International Public-Sector Accounting Standards, International Accounting Standards and International Financial Reporting Standards as issued by the International Accounting Standards Board.

7.3 Significant accounting policies

The program is responsible for attaining the internal financial controls and value for money during the implementation of program activities. The accounting policies adopted in the preparation of financial report are set out below. These policies have been consistently applied to all years presented, unless otherwise stated.

Note 1: Basis of accounting: the financial annual report is prepared on a cash receipts and disbursements basis and whereby fund is recognized on receipt of cash and expenditure recognized on the disbursement of cash or cash equivalents except for the sub-recipient or sub-office Pemba cash disbursed is recognized as advance.

Note 2: Disbursements/receipts: this represents the funds that are received by ZAMEP from Revolutionary government of Zanzibar and development partners under the grant agreement during the period of July 1, 2021, to June 30, 2022, which include direct and indirect disbursement.

Note 3: Program advances received: funds received by principal recipient and fund transferred to sub-recipient are both recorded as advance obligations until they are spent by the program, at which they are recognized as expenditure.



Note 4: Government funding: funds from the government for program activities is recognized as fund received except indirect costs like staff salaries, electricity, and infrastructures.

Note 5: Functional and presentation currency: transactions in Tanzania shillings are translated to US dollars at the exchange rate applicable at the spot rates applied by bank on the date of transfer of the funds from US Dollar bank account to TZS bank account. For the purpose of this report, balances denominated in Tanzania shillings have been translated to US dollars based on the weighted average exchange rate as reported in Bank of Tanzania during the period which stood at TZS 2,309.65 / US\$.

Balances at the year-end denominated in local currencies are translated using the spot exchange rates at the end of reporting. The Foreign exchange gains or losses resulting from translation of local currency to foreign currency.

7.4 Budget Allocation

Financial resources for malaria interventions came from two major sources: the government, and donors. As shown in table 17, the donors are the major financier of malaria interventions.

The government and donors' contribution in the program has gradually increased compared to the FY2020/2021. The total of US\$ 5.6 million was approved for the FY 2021/2022 to cover various malaria activities and C19RM activities compared to US\$ 4.2 million for the FY 2020/2021. This allocation shows the increasing of fund by US\$ 1.4 million. The increase in the allocation indicates the government and donors' commitment to fight against malaria and C19RM.

Table 10 : Budget allocation

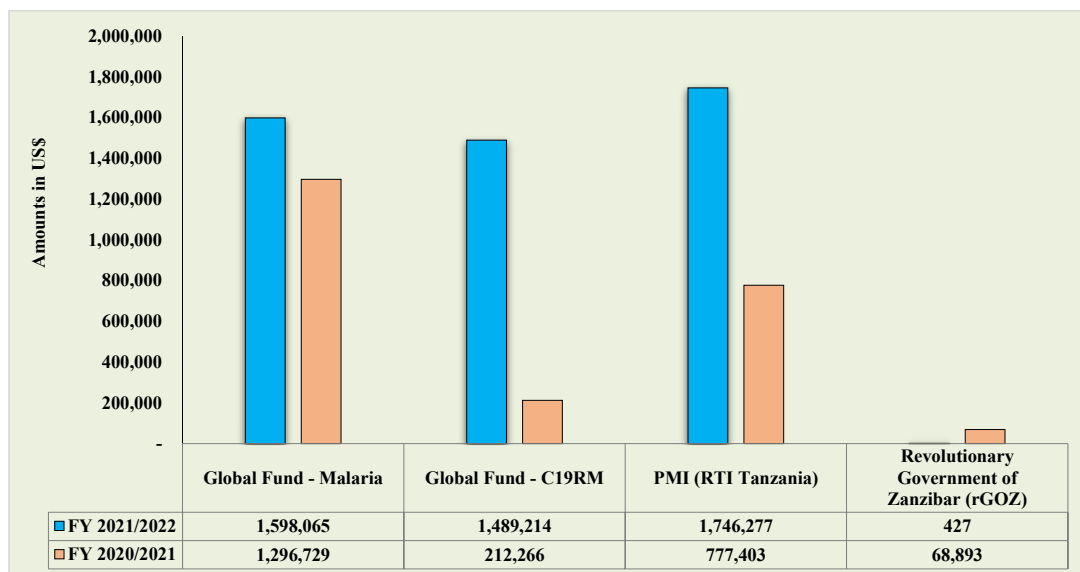
By Module	Global Fund	PMI	RGOZ	Total
	Amount in US\$			
Case management	336,336	429,610	59,986	825,932
Vector control	293,436	1,026,210	201,329	1,520,976
Program management	261,890	190,076	47,626	499,592
RSSH: Health management information systems and M&E	154,441	601,739	-	756,180
RSSH: Community systems strengthening	31,141	-	-	31,141
RSSH: Health products management systems	136,968	-	-	136,968
COVID-19	1,855,196	-	-	1,855,196
Total	3,069,408	2,247,636	308,941	5,625,985

Source: ZAMEP extract data 2022

7.5 Financial disbursement

Figure 19 represent the total funds received from each donor. During 2021/2022 the total fund received was US\$ 4.8 million. This shows an increase of funds to a total of US\$ 2.5 million in comparison to 2020/2021, During the FY the Global Fund provided a total of US\$ 3.087 million, of which US\$ 1.59 million was for Malaria and US\$ 1.48 million for C19RM, the increase of funds was equivalent to US\$ 1.5 million. Similarly, funds received from PMI/USAID through RTI had a similar increased trend from US\$ 0.7 million in 2020/2021 to US\$ 1.7 million in 2021/2022 equivalent to an increase of US\$ 0.9 million. In addition, funds from the RGOZ significantly reduced from US\$ 68,893 in 2020/2021 to US\$ 427 in 2021/2022 equivalent to 0.001% of the total fund allocated for this financial year.

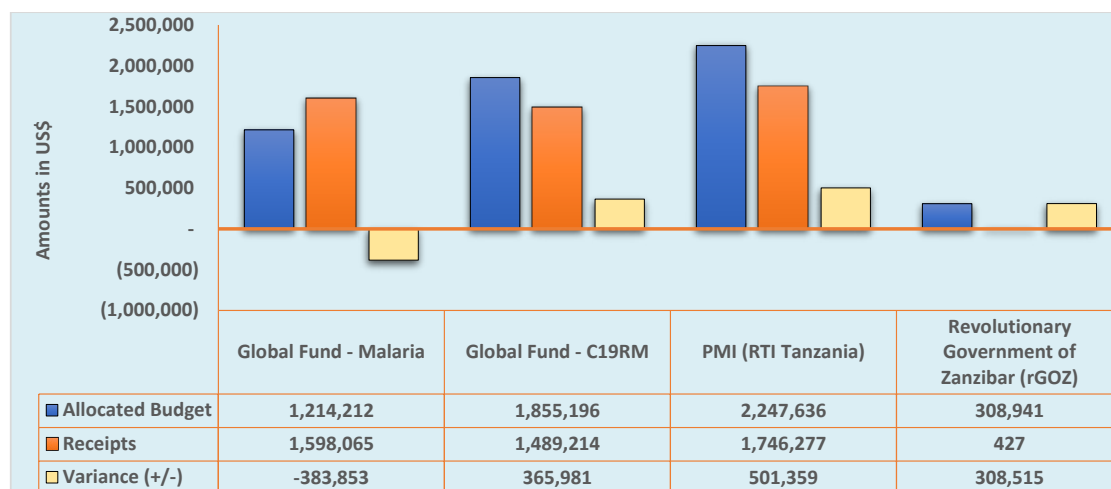
Figure 12 : Funds Receipts



Source: ZAMEP extract data 2022

7.6 Allocated budget and Receipts

Figure 20 represents the amount allocated compared to the amount disbursed. The data shows the fund disbursed from global fund for malaria activities exceed the budget allocated for US\$ 0.017 million. This is due to PPM commodities (LLINs) which were planned in FY 2020/2021 but consignment delivered and paid in FY 2021/2022, for the part of C19RM and PMI (RTI Tanzania) the fund was not disbursed all due to delay of C19RM commodities and therefore other activities were carried to the next financial reporting period. The RGOZ disbursed only 0.14% of the total fund allocated for the FY 2021/2022.

Figure 13 : Allocated Budget and Receipts

Source: ZAMEP extract data 2022

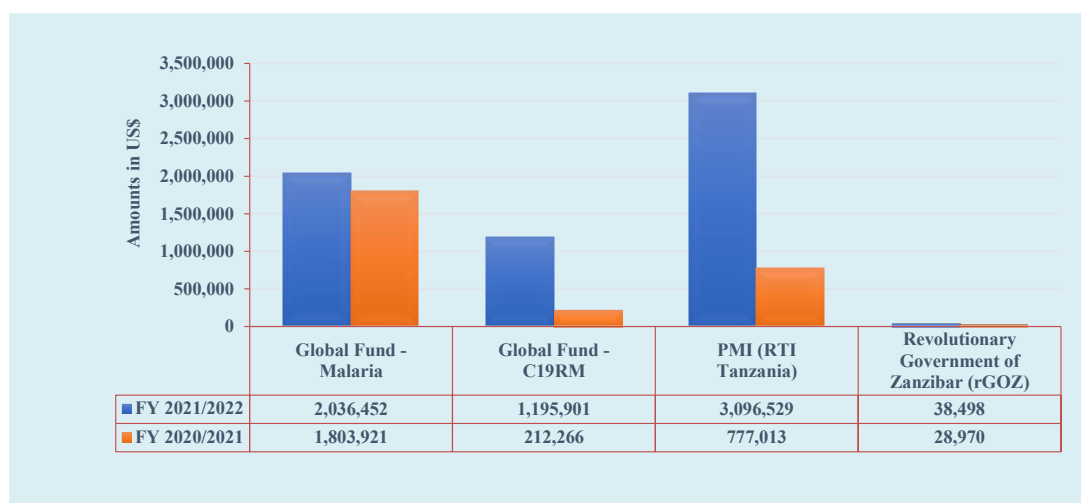
7.7 Program Expenditures

Expenditure represents amounts of costs incurred by PR and SRs during the financial reporting period, which is denominated in USD. The main expenditures for the funds received were used to cover the malaria interventions as follows.

- Entomological Monitoring
- Insecticide Treated Nets
- Community systems strengthening Surveillance, Monitoring and Evaluation
- Social Behavior Change
- Health products management systems which include procurement of pharmaceutical and non-pharmaceutical products.
- Covid-19 Interventions

Figure 14 shows the total program expenditures (PR & SR's) for FY2021/2022 was US\$ 5.03 million which was higher comparing to FY2020/2021 which was US\$ 2.8 million. Moreover, during this reporting period the total of 0.9 and 1.4 million were used to procure C19RM and Malaria commodities respectively though Pool Procurement Mechanism. The general trend of total expenditure showed an increase by US\$ 2.1 million. The increase was mainly attributed by additional C19RM fund. Looking at each component, funds from Global Fund and PMI (RTI Tanzania) increased by US\$ 2.1 million and US\$ 0.28 million from 2019/2020 to 2020/2021 respectively. The spending from RGOZ increased from \$ 28,970 in 2020/2021 to US\$ 38,498 in 2021/2022. However, the fund used for this reporting period was the commitment from FY 2020/2021.

Figure 14 : Malaria operating expenditure



Source: ZAMEP extract data 2022

7.8 Financial performance

The comparability of financial performance for the FY 2021/2022 between fund budgeted and funds spent by each program module has been presented in percentage in and graph 22 shows Case management has higher in spending with a 120% burning rate followed by Vector Control which has a burning rate of 115% and lower is COVID-19 with 64%. The significant variance for overspending was due to delayed payments to the commitment's procurement of FY2020/2021 and hence reported as expenditures in this financial period after delivered to PR.

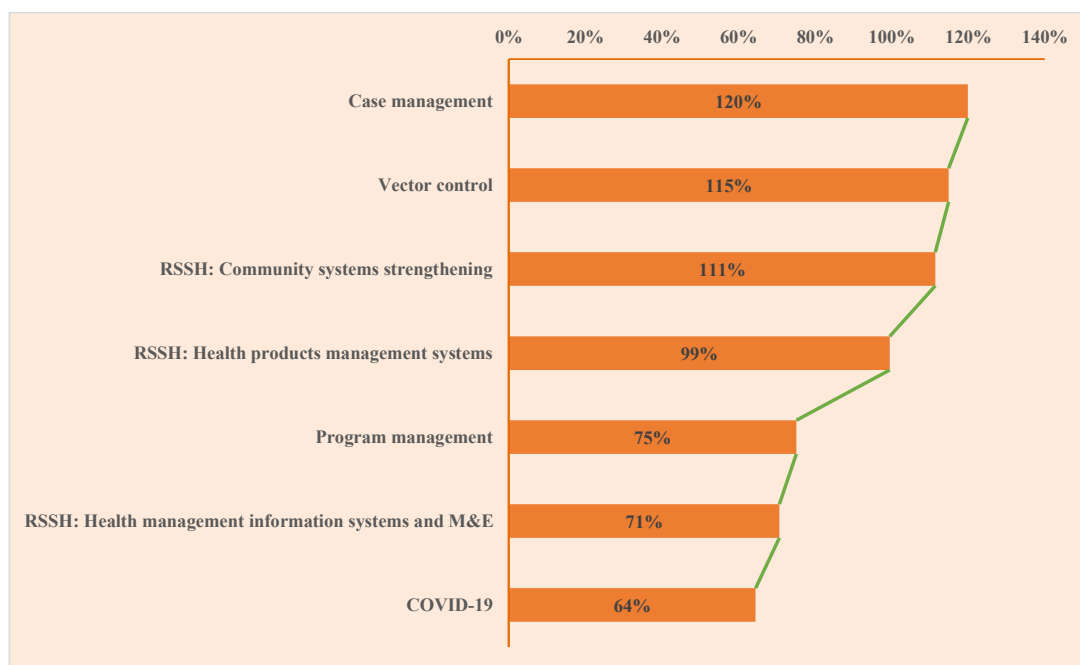
Under absorption from C-19RM was caused by procurement of oxygen plant, the tender was delayed due to insufficient fund to cover procurement cost. The overall burning rate for all interventions is around 89% which indicates that some of the activities from various thematic areas have not been implemented at the end of financial period, the unimplemented activities have been committed for the next financial year.

Table 11: Allocation budget and expenditure

Intervention	Budget in \$	Expenditure in \$	Burning Rate
Case management	825,932	990,094	120%
Vector control	1,520,976	1,746,920	115%
RSSH: Community systems strengthening	31,141	34,698	111%
RSSH: Health products management systems	136,968	136,240	99%
Program management	499,592	375,068	75%
RSSH: Health management information systems and M&E	756,180	534,506	71%
COVID-19	1,855,196	1,195,901	64%
Total	5,625,985	5,013,426	89%

The total expenditure reported including the indirect payment (PPM) for the procurement of LLINs, Malaria Rapid Diagnostic Test kit, IRS, COVID-19 Diagnostic Tests, COVID-19 PPE, COVID-19 Oxygen Equipment and Artesunate plus Amodiaquine, Laboratory consumables, Equipment and Supplies for Entomological monitoring, motorcycles, and establishment of shokishoki system. The total cost of all PPM was US\$ 2.4 million.

Figure 15: Absorption Capacity



Source: ZAMEP extract data 2022

7.9 Financial Commitments

A total of USD 408,275 which was not spent at the end of June 2022 were commitment for the following activities.

- 1) Procurement of oxygen plant facilities and installation
- 2) Support council malaria surveillance officers to conduct case investigation
- 3) Payment of internet bandwidth to health facilities for data entry into DHIS2
- 4) Hosting of DHIS2 data base in cloud server (hosting of database, domain name etc.)
- 5) Orientation of district surveillance officers and health facilities supervisors for EBS



- 6) Conduct 2-weeks molecular biology training using biorad PCR machine at Unguja and PHL Pemba
- 7) Support the reviews/ update, printing, and distribution of COVID-19 SoPs
- 8) Conduct MEEDS, IDSR & MCN feedback Meeting

The implementation of the commitment will take place from July 2022, and they will be reported in the coming financial year.

7.10 Audit Status

ZAMEP has been audited its book of Accounts and Financial Statement by CAG for the period of FY2020/2021 and rated as an unqualified opinion with minor issues to be rectified. Audit for the financial year 2021/2022 (July to June) is in the final stage, the final report from the Controller and Auditor General (CAG) will be out in November 2022.



CHAPTER EIGHT

MAJOR OPERATIONAL CHALLENGES AND RECOMMENDATIONS

8.1 Operational Challenges

1. Irrational prescription of antimalarial particularly among private health facilities.
2. Use of multiple brands of mRDTs in Private health facilities, some of which having low performance standards leading low detection of malaria cases.
3. Use of outdated tools that captures only indoor biting data while most of the mosquitoes bites are happening outdoors now a days, this compromises ITNs and IRS performance
4. High number of imported malaria case from nearby regions with high incidence of malaria
5. Frequent transfer of the trained health workers affecting performance of some core malaria interventions

8.2 Recommendations

1. Strengthen supervision and on-job trainings on diagnosis and rational treatment of malaria among health care workers particularly in Private Hospitals (Case Management)
2. Provide adequate support to health facility staff to familiarize with new IDSR reporting system during this transition period from MEEDS to IDSR systems (SME/HMIS)
3. Revise entomological and surveillance guidelines to capture more information based on current malaria situation and WHO standards (Program management/Entomology Unit)
4. Mobilize more resources to maximize implementation of larva source management activities as core interventions for malaria elimination (Vector control /Program Management)
5. Finalize and fasten implementation of multispectral guidelines to address issues of importation of malaria cases in Zanzibar
6. Strengthen genotypic analysis of malaria parasite to enhance species identification and specification